# FREQUENCY RECOVERY AND THE LIMITS OF READOUT EXPLANATIONS IN ARBITRARY-SCALE SUPER-RESOLUTION

*Internal scientific review draft*

Author information to be confirmed before submission


## ABSTRACT

In arbitrary-scale super-resolution, cell inputs are often motivated by the size of a target pixel, but frequency ordering alone does not establish what this readout computes. We evaluate recovery in low-resolution Nyquist units and distinguish paired training ablations from fixed-weight cell interventions. Normalisation reduces cross-scale variation in all nine primary model–dataset groups and all 27 groups including additional seeds; the original strict collapse criterion still fails. Fixing image weights reduces variation on Set14 but increases it on Urban100: pooled consistency does not imply image-wise consistency. Coarse readout-loss ordering replicates in four architecture–seed groups, yet a footprint score has exactly the ranks of band index and scale averaging fails function similarity in every group. A separately calibrated, observation-only forecast also misses harmful interventions and does not improve on a magnitude/spectrum baseline. These results support a frequency-based evaluation coordinate while delimiting the tested readout interpretations and instance-level guidance.


***Index Terms—*** Arbitrary-scale super-resolution, frequency recovery, cell conditioning, readout diagnostics

## 1. INTRODUCTION AND RELATED WORK

Arbitrary-scale super-resolution (ASISR) predicts image values at continuous coordinates. LIIF supplies a target cell size to a local decoder [1]; LTE introduces cell-dependent phase estimation [2]. Such conditioning can affect high-frequency errors, but two questions are easily conflated: does a readout comparison exhibit a frequency trend, and does that trend explain the computation or predict when the readout is harmful? We study this distinction rather than introducing another decoder.

Both the positive and negative roles of cell inputs have precedents. LIIF already varies cell size at fixed queries and reports small large-scale disadvantages in its ablation. IPE-LIIF integrates positional features over a pixel and reports cases where an additional cell input hurts [3]. Thera provides analytic Gaussian filtering and discusses a discrepancy between its theoretical filter and the bicubic training degradation [4]. Thus cell intervention, integrated encoding, harmful cell effects and filter mismatch are not new principles here. Integration of intermediate features also need not equal integration of a nonlinear decoder's final RGB output.

Frequency-resolved SR analysis likewise predates this work. HyRA studies linear/nonlinear responses and frequency-spectrum similarity [5], while texture-based error analysis goes beyond aggregate PSNR [6]. Active sinusoidal probing also diagnoses degradation mismatch [7]; Nyquist-normalised encoding and spectral consistency appear in spatiotemporal forecasting [8]. Our fixed-checkpoint diagnostic normalises reconstruction error by target signal energy and compares frequencies in LR Nyquist units. Architecture/recipe effects are studied by a wider continuous-SR retraining benchmark [9]; our fixed checkpoints do not resolve that benchmark's general model comparison. Generalized SURE [10] supplies risk estimates under stated assumptions, whereas the bicubic proxy examined here has no unbiased-risk guarantee.

Our contributions are two empirical tests with explicit simple controls. First, we quantify the reduction in scale variation obtained by Nyquist normalisation and distinguish image-wise variation from energy-weight migration in pooled curves. Second, we distinguish coarse ordering, quantitative prediction, function equivalence and harmful-intervention detection. Replicated ordering coexists with failed stronger tests: it is not evidence for a shared integral readout or a reliable instance-level decision. The contribution is this controlled evidence and its limits, not the first observation of harmful conditioning or a new risk theorem.

## 2. MEASUREMENT AND COMPARISON DESIGN

### 2.1. Separate sampling, target and conditioning

Let $y = Ax$ be the LR observation of a continuous signal, $Q_h x$ the desired measurement at a query coordinate $q$, and $c$ the cell supplied to $F_\theta(y, q, c)$. Changing $A$, changing $Q_h$ and changing $c$ are different interventions. Assigning $c = h$ does not prove that $F_\theta$ implements $Q_h$. Natural-image experiments use discrete HR targets and bicubic antialiased LR preparation, not certified box averages of a known field. Pixel-footprint arguments therefore motivate tests but do not define ground truth in those experiments.

The evaluated LIIF/LTE/SC-INR variants inherit an EDSR-baseline encoder, DIV2K training [11], 1000 epochs and train-

ing scales $\times 1$–$\times 4$. SC-INR denotes the examined implementation with sinc-modulated Fourier features; its phase/readout is not identified with LTE's. A trained full/no-readout pair changes weights as well as the readout path. A fixed-weight native/zero-cell comparison instead changes only conditioning, and zero cell need not be in training support. The LTE snapshot used in frozen-weight probes omits the extrapolation cell clamp described in the original paper; conclusions concern these checkpoints, not all published LTE implementations.

### 2.2. Frequency recovery and scale variation

For frequency band $b$, write $S_{ib} = \|\Pi_b(x_i - \bar{x}_i)\|^2$ and $E_{ib} = \|\Pi_b(x_i - \hat{x}_i)\|^2$. Recovery relative to the centred zero reconstruction is

$$R_b = 1 - \frac{\sum_i E_{ib}}{\sum_i S_{ib}}. \tag{1}$$

It can be negative and is not the equal-image mean of $1 - E_{ib}/S_{ib}$. The frequency study (L1) uses the RGB mean, unwindowed FFTs on centre $240 \times 240$ crops, and Chebyshev frequency $\nu = \max(|\nu_x|, |\nu_y|)$. It covers 14/30/30 images from Set14/BSD100/Urban100 [12, 13, 14]. Seven bins have edges $.25, .5, \ldots, 2$ in $u = 2s\nu$ for $s = 2, 3, 4$; the LR Nyquist boundary is $u = 1$. The absolute-frequency control fixes the same bins at $\nu = u/6$. Variation is

$$D = \mathrm{median}_b\{\max_s R_b(s) - \min_s R_b(s)\}. \tag{2}$$

This compares two evaluation coordinates with different bin supports; it does not prove equality of reconstructed functions.

### 2.3. Readout losses and observation-only forecasts

The paired training ablation and its replication (F1/G1) use centre HR256, 13/30/30 images, the RGB mean and three HR frequency bands with cutoffs $1/8, 1/4, 1/2$. Their paired loss is $L_b = R_b^{\mathrm{full}} - R_b^{\mathrm{ablated}}$; positive means the full model helps. The footprint score $(.00006, .00552, .07235)$ and its replication criteria were frozen before seed2/3 evaluation. Its three strictly increasing values have exactly the same tied ranks as band index $(1, 2, 3)$. The scale-average test (L2), with HR240 and 14/30/30 images, averages the full model over 31 conditioning scales $s' \in \{1, 1.1, \ldots, 4\}$ at fixed features and coordinates. It tests the loss forecast and whether this averaged output resembles the trained ablation. This operational average is not a posterior over scale; HR truth enters its recovery score.

For the separate observation-only test, let $F_c, F_0$ be fixed-weight native/zero-cell RGB outputs, $d = F_c - F_0$, and $B(y)$ bicubic upsampling. True gain and the proxy forecast are

$$G = \mathrm{MSE}(F_0, t) - \mathrm{MSE}(F_c, t), \tag{3}$$

$$\hat{G} = 2\langle B(y) - F_0, d\rangle - \|d\|^2, \tag{4}$$

$$G - \hat{G} = 2\langle t - B(y), d\rangle. \tag{5}$$

**Table 1**. Primary $D$ in LR-Nyquist units. All three Set14 entries fail the original .05 tolerance.

| Model | Set14 | BSD100 | Urban100 |
|---|---|---|---|
| LIIF | .0586 | .0154 | .0165 |
| LTE | .0548 | .0149 | .0190 |
| SC-INR | .0605 | .0128 | .0147 |

Inner products and norms are pixel averages, optionally after band projection. Equation (5) is an elementary error identity: unknown target/proxy error aligned with the correction changes its gain. It is not an observation-only guarantee.

Calibration uses DIV2K train0001–0032; evaluation uses valid0825–0836, centre HR128, $\times 2/4$, no added noise, raw RGB outputs and shave4. These interventions use one seed1 checkpoint per architecture. One forecast uses Eq. (4); another adds proxy alignment to a calibrated ridge with scale/band indicators and observable change energy, output/proxy energies and LR gradient/spectrum summaries. The matched ridge baseline omits only alignment; standardisation is calibration-only, with fixed ridge penalty 1 and an unpenalised intercept. Band/scale median, energy-only, always-enable and always-disable controls are also retained. All coefficients, forecasts and decisions were frozen before evaluation scoring, with no test tuning. The evaluation images were not used to fit these forecasts; their use in the original model validation is not established. Forecasting requires both outputs and is not a pre-inference shortcut.

## 3. RESULTS

### 3.1. A useful coordinate, not a universal collapse law

Nyquist normalisation reduces $D$ in all nine primary groups: .013–.061 versus .31–.46 in absolute frequency. Figure 1 illustrates one successful dataset and the Set14 exception; Table 1 retains all primary results. The original criterion ($D \leq .05$ in at least eight groups) fails, with six of nine passing. This difference cannot be replaced by a post-hoc relaxed tolerance.

A fixed descriptive sensitivity analysis reuses saved ring energies, including two further seeds. Nyquist normalisation reduces $D$ in all 27 model–dataset–seed groups for every tested setting: the original domain, below/above Nyquist subdomains, each leave-one-scale-out pair, and pooled/equal-image aggregation (24 settings). These are dependent comparisons, not 648 independent replications. In contrast, the strict .05 tolerance passes 19/27 with original pooling and 27/27 with equal-image averaging on the original domain. Thus the qualitative normalisation benefit is stable here, while a strong collapse statement depends on how images are weighted. The primary failure is retained.

To examine this dependence, let $R_{isb} = 1 - E_{isb}/S_{isb}$ and $w_{isb} = S_{isb}/\sum_j S_{jsb}$. The pooled score is $\sum_i w_{isb} R_{isb}$.

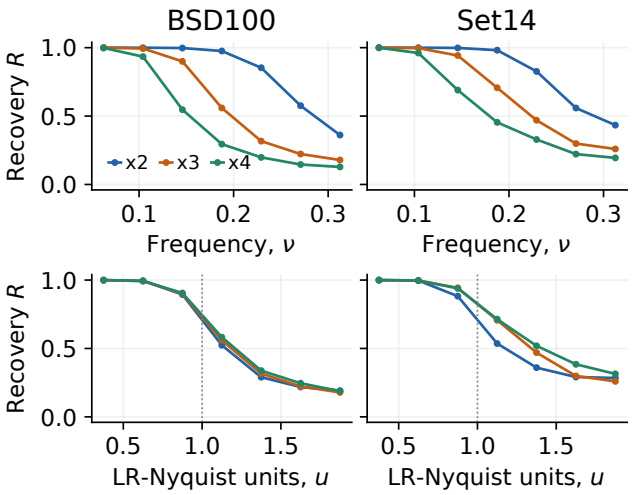



**Fig. 1**. SC-INR seed1: absolute-frequency (top) and LR-Nyquist (bottom) recovery. All primary datasets enter Table 1; these panels show consistency on BSD100 and residual scale differences on Set14. Recovery uses pooled energies, not a mean of image-wise scores.

A retrospective diagnostic holds weights fixed at $\bar{w}_{ib} = \frac{1}{3}\sum_s w_{isb}$, retaining all images and predictions. This reduces $D$ in every Set14 group, from .0493–.0616 to .0227–.0294, but increases it in every Urban100 group, from .0147–.0202 to .0293–.0538; two Urban100 groups then exceed .05. Weight migration can therefore amplify or conceal the measured variation. It is not an independently identified physical cause: the same normalised bin uses different HR frequencies at each scale, and $S$ enters both the weights and individual scores.

Individual scores also remain dispersed: applying the same .05 tolerance descriptively gives only 7–9 of 14 images per Set14 group, 23–26 of 30 in BSD100, and 14–17 of 30 in Urban100. All nine original Urban100 pooled curves pass, despite these image-wise exceptions. These are dependent retrospective checks, not a new image-wise pre-registered criterion. Fixed weights use HR spectra and are not an observation-only selector; neither reweighting nor deleting an image repairs the original failed criterion.

### 3.2. Replicated ordering does not identify a readout

Table 2 joins the four additional-seed tests. G1 meets its $\rho \geq .5$ coarse-ordering threshold in all four groups; the high-band loss exceeds the middle-band loss in 34/36 dataset–scale settings. Equal-image aggregation gives 36/36 and correlations $.641, .943, .646, .856$, preserving that weak conclusion. Per-image recoveries permit this sensitivity check, but missing original per-image energies prevent independent re-pooling of G1. The footprint formula has no rank advantage over band index; multiplying by the proposed recovery factor lowers correlation in all four groups.

**Table 2**. Paired training ablations. G1: loss correlation with band index (identical to footprint ranks). L2: scale-average loss correlation, median amplitude ratio on $L > .005$, and function similarity fraction.

| Group | G1 $\rho$ | L2 $\rho$ | Ratio | Similarity |
|---|---|---|---|---|
| LTE seed2 | .577 | .712 | .823 | .770 |
| SC seed2 | .943 | .755 | .581 | .689 |
| LTE seed3 | .676 | .954 | .833 | .748 |
| SC seed3 | .885 | .700 | .337 | .653 |

**Table 3**. SC harmful-band detection, 14 negatives/58 positives. TP is detected harm; FP is beneficial output incorrectly flagged harmful; BA is balanced accuracy.

| Forecast | TP | FP | BA |
|---|---|---|---|
| Magnitude/spectrum ridge | 6 | 15 | .585 |
| Ridge + alignment | 6 | 17 | .568 |
| Bicubic proxy | 4 | 15 | .514 |

L2 ordering ($\rho \geq .7$) passes 4/4 groups and its amplitude interval $[.5, 2]$ passes 3/4. However, similarity (the ablation being closer to the averaged than the full output) never reaches the frozen .8 threshold. L2 exceeds the simpler footprint correlation by .05 in only 2/4 groups, both LTE, below the required 3/4. Different crops/colour pooling mean G1 and L2 are interpreted within their own protocols; L2 superiority uses its own recomputed footprint comparator, not the G1 column. High correlation does not establish function equivalence or uniform quantitative superiority.

The intervention distinction is consequential even for the same image. On barbara at $\times 2$ in the high band, the trained SC full/ablation pair has gain $-.193296$, while the fixed-weight native/zero-cell pair has gain $+.009733$. These different comparisons cannot be pooled as measurements of one mechanism. Known-field target/hint probes also contain both gain signs within every band; their synthetic and unsupported-query scope prevents a general natural-image causal claim.

### 3.3. The tested forecasts miss harmful interventions

For SC, 14 of 72 evaluation band outcomes are harmful ($G < -10^{-8}$), spanning nine images. Table 3 shows that alignment does not add reliable discrimination over the matched magnitude/spectrum baseline. The paired 12-image-cluster bootstrap 95% interval for its balanced-accuracy improvement is $[-.167, .143]$. LTE supplies just one harmful band outcome, so its high apparent balanced accuracy cannot establish reliable negative detection. Bands/scales share images; bootstrap intervals are marginal, not joint 95% guarantees.

The original success gate requires at least 12 negatives on

at least six images, recall $\geq .75$, precision $\geq .6$, false-positive rate $\leq .25$, and an advantage over every simple baseline (balanced accuracy by $.05$ and a positive paired bootstrap lower bound). A separate full-RGB policy must reduce MSE relative to every control with a positive lower bound. Both forecasts choose native output iff the full-RGB forecast is positive, else zero-cell. Both fail the combined gate on both models. SC proxy policy reduces pooled MSE by $1.31 \times 10^{-5}$ relative to always enabling, but its interval $[-6.81 \times 10^{-6}, 4.78 \times 10^{-5}]$ crosses zero and mean PSNR falls $.0047$ dB. The alignment policy is slightly worse in MSE. These are not adaptive benefits hidden by the classification threshold.

Saved-output checks with shave0/4/8 retain SC harm in 16/14/14 original band outcomes and 6/5/5 full-RGB outcomes; LTE has 1/1/1 harmful bands and no harmful RGB outcome. Two fixed alternative band partitions give 10–16 harmful SC outcomes; harm is not confined to the original cutoffs. These post-evaluation sensitivity checks do not refit forecasts or claim new confirmation. The original fixed RGB policies still lack a reliable SC improvement at every shave; the LTE proxy policy is harmful throughout.

## 4. IMPLICATIONS AND LIMITS

LR-Nyquist frequency is useful for comparing recovery across scales, but pooled consistency also depends on image weights and can hide image-wise exceptions. Reweighting has opposite effects across datasets; it is not a universal correction. Readout evaluation should report signed gains and simple band controls; rank correlation with a three-valued predictor cannot validate its quantitative form. Training ablations and fixed-weight conditioning answer different questions and can give opposite signs. Within these checkpoints, the tested proxy adds no reliable harmful-intervention guidance; this does not exclude better proxies or mechanisms.

Scope is limited to the inherited recipes, crops, scales and three seeds. There is no fair performance benchmark against all integrated-readout methods, no certified physical HR footprint, and no universal decoder limit. Zero-cell/synthetic queries introduce support differences. All primary failures and saved outputs are retained; independent numerical audits verify scoring, not a new model implementation. The new natural forecast test takes 224.19 seconds on one CPU thread, with no GPU or SR training; offline robustness adds no inference. Its cost includes both native and zero-cell predictions. These results establish a controlled empirical baseline; a predictive rule needs separate validation.

## 5. CONCLUSION

Frequency normalisation yields a robust reduction in measured scale variation, whereas pooled consistency need not hold image-wise. Coarse readout ordering replicates, but footprint ranking has no advantage over band index and does not validate the stronger function or harmful-benefit forecasts examined here. Separating these evidence levels makes ASISR readout claims testable without treating correlations as mechanisms.

## 6. ACKNOWLEDGMENTS

Claude and OpenAI Codex assisted with text, analysis code, figures and record checks in Sections 1–5. Results were checked against stored records; scoring audits are not model reimplementations. Human author review and responsibility are required before submission.

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